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西交利物浦大学

Department of Electrical and Electronic Engineering

MEC202 Industrial Awareness and Group Project

Group Project Report

PDR LOCATION APP

Group 34

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1. Introduction

Background:

Pedestrian Dead Reckoning (PDR) is a technique commonly employed in indoor positioning systems for monitoring the movements of individuals in the absence of GPS. This technique relies on the use of sensors such as accelerometers, gyroscopes, and magnetometers embedded in a wearable device or smartphone to estimate the user's location and movement[1]. PDR is based on the principle that the user's movement can be tracked by measuring changes in acceleration and orientation of their device. The system utilizes this data in combination with a map of the indoor environment to estimate the user's location and track their movements as they navigate within the facility. In recent years, PDR has gained popularity due to the increasing adoption of smartphones and wearable devices. It has numerous applications, including indoor navigation, fitness tracking, and location-based services. Given the common challenge of getting lost and being late for class among students, particularly freshmen, our group developed a PDR location app that utilizes the PDR algorithm and the accelerometer, magnetometer, and gyroscope sensors integrated into mobile phones. The aim of the app is to assist students in addressing this challenge.

1.1 User survey

In order to know the potential user demand and market feasibility of our product before designing and developing the product, we made a questionnaire and opened it to the students of the school. The survey was conducted across the university, with a particular focus on freshmen.
Considering the convenience of data statistics in the later period of the questionnaire and clearer analysis of results, we only use a platform named Wenjuxing.

The questionnaire:
We are developing an indoor positioning and navigation software based on PDR technology. To do this we need to collect data about the intentions of potential users. May I have a moment of your time to finish this questionnaire? Your answers and suggestions are very important to us. Thank you for your help.

1. How often do you use online maps, such as Amap?
   A. Often B. Somtimes C. Occasionally D. Never

2. Will you use online maps in your school?
   A.Yes B. No

3. In which of the following scenarios would you use an online map in your school?(multiple options can be selected)
   A. Looking for teaching buildings on campus
   B. Look for classrooms in the teaching building
   C. Other:_____

4. How do you feel about using online maps in your school or building?
   A. Very good, always find the destination quickly
   B. In general, occasionally get lost, but always find the destination
   C. Poor, often lost
   D. It's bad. I can never find my way

5. Has the online map ever been unavailable due to poor signal, especially in the building?
   A. Yes B. No

6. Do you think this has greatly influenced your actions on campus?
   A. Yes B. No

7. Do you want a mapping app that works in places where GPS signals are weak?
   A. Yes B. No

8. What would you recommend for such a piece of software?
   _____

The results:

1. How often do you use online maps, such as Amap?

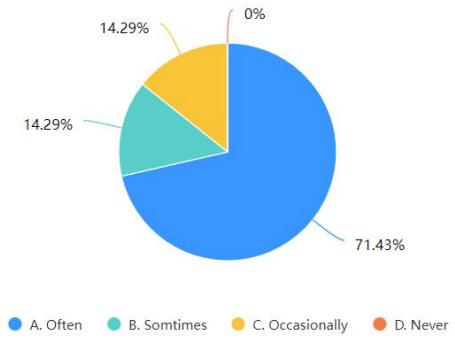


Fig 1. The pie chart for question 1

2. Will you use online maps in your school?

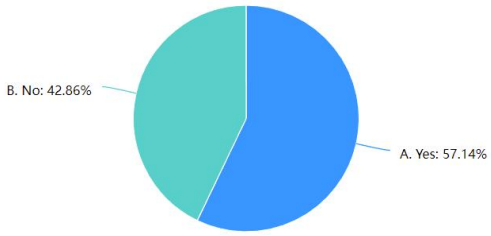


Fig 2. The pie chart for question 2

3. In which of the following scenarios would you use an online map in your school?(multiple options can be selected)

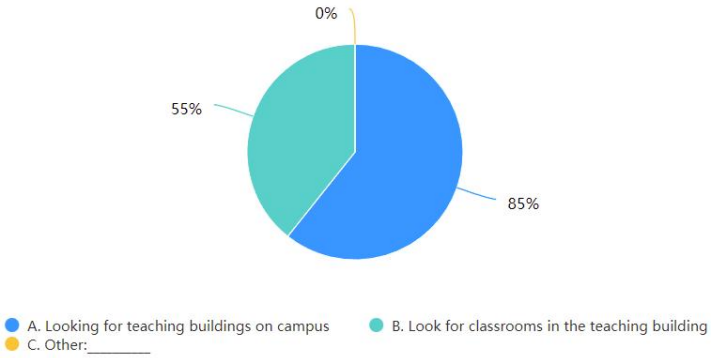


Fig 3. The pie chart for question 3

4. How do you feel about using online maps in your school or building?

Fig 4. The pie chart for question 4

5. Has the online map ever been unavailable due to poor signal, especially in the building?

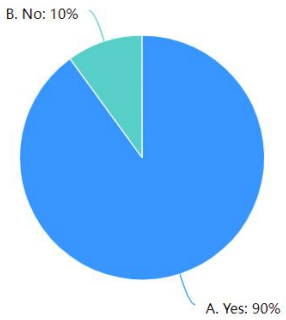


Fig 5. The pie chart for question 5

6. Do you think this has greatly influenced your actions on campus?

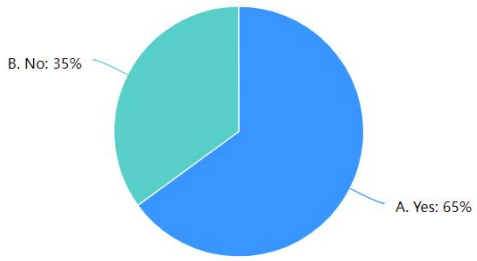


Fig 6. The pie chart for question 6

7. Do you want a mapping app that works in places where GPS signals are weak?

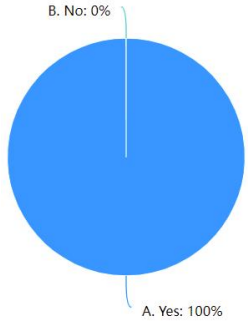


Fig 7. The pie chart for question 7

8. What would you recommend for such a piece of software?

As for the feedback of this problem, we drew a conclusion through analysis: the biggest demand of potential users for this product is that the operation of the software must be convenient and fast, the positioning should be real-time and accurate, and the location of each classroom should be accurately marked on the map.

In conclusion, we received 105 valid questionnaires, most of which indicated that they would use online maps regularly, and 60 indicated that they would use online maps in school. Although almost all of them said that the current online map is barely adequate to find buildings and classrooms on campus, they all agreed that the online map has a poor use experience due to poor signal in the buildings, affecting the movement on campus. To our surprise, all of them said they wanted a mapping app that could work in places where GPS signals were weak.

Since freshmen are not very familiar with the road of the school and the structure of the teaching building, we specially selected several domestic and international students to interview them. Here are three typical responses.

Intervewer: It has been several months since you came to the new school. Where do you spend most of your time attending classes or studying by yourself? Do you often struggle to find teaching buildings and classrooms on the way? Do you ask people around you if you can't find the way? Are you interested in our products based on PDR technology?

Student A(Year 1 Domestic): Most of my classes are in the foundation building, and it's easy to find the way to the foundation building. But because of its symmetrical

structure, I often forget whether I am on the east or west side of the building. If I can't find the way, I will ask the upperclassmen around me. I am very interested in this new product and I hope to use it early so that I can find my classroom quickly.

Student B(Year 1 From Indonesia): Most of my classes are in the Science building(including SA, SB, SC and SD), but because of the epidemic, many of the entrances and exits of the science building were closed last year, so it was difficult for me to find the nearest entrance and exit to the classroom. In addition, the science building is connected by four separate teaching buildings, the structure inside is like a maze, I often get lost in the teaching building, even can't find the exit. Since my Chinese is not very good and I'm a little shy, I don't ask people around me for directions. I tried to use the wechat mini program Xipu Map, but due to the weak signal in the teaching building, coupled with the inaccurate map and positioning, it was still difficult for me to find the classroom I wanted to go to. So I wish I could use the product earlier so I could quickly find the classroom on my own.

Student C(Year 1 Domestic): I often study in the central building, because the library is in it, and I think the academic atmosphere there is very good. At first, I often couldn't find the bathroom when I was studying, or couldn't find my seat when I came out of the bathroom. As I had to be absolutely quiet in the library, I couldn't ask people around me for directions. Although this situation has improved in the future, I am still looking forward to a product that can help me find my way around the library.

1.2 Existing similar industrial products

The contemporary market offers a wide range of products that bear similarity to our PDR location app. Nonetheless, these products often suffer from design inadequacies that constrain their efficacy. For instance, Amap is a product that prides itself on providing online, real-time location tracking and high adaptability. While it excels in accurately locating users via GPS, it exhibits noticeable limitations such as the inability to furnish indoor location information, a dearth of customization options, and inefficacy in locating users in areas with weak GPS signals. For instance, a neophyte college student may encounter difficulties navigating to a classroom within a vast, unfamiliar building, where Amap may not provide expedient guidance.

An additional instance is the Xipu Campus Map, a location app that was created by students at XJTLU. The app caters to the individualized requirements of XJTLU's students by streamlining the process of locating unfamiliar buildings. Nonetheless,

similar to Amap, the Xipu Campus Map encounters limitations associated with its utilization of GPS location tracking. Specifically, it may be unable to provide efficient guidance in areas where the GPS signal strength is feeble within campus buildings and may be incapable of presenting detailed classroom information to facilitate precise classroom navigation by students.

To overcome these limitations, our PDR location app employs sophisticated algorithms and sensor fusion techniques to furnish accurate location data, even in indoor areas or those with suboptimal GPS signals. Moreover, our app provides adaptable settings that cater to the particular needs and demands of our users, presenting a comprehensive solution for location tracking requirements within industrial settings.

1.3 Summary of specific customer needs and design requirements

1. The system should provide accurate, reliable and real-time indoor positioning information, enabling users to achieve effective positioning in various indoor environments.
2. The solution should be user-friendly, with an intuitive interface that makes it easy for users to understand their location data.
3. The system should be scalable, allowing for future improvements, integration with other technologies, and compatibility with a wide range of devices.
4. The system should maintain user privacy by securely processing and storing personal data collected during the positioning process.
5. The solution should offer customization options, such as adjustable settings for different environments or user preferences, to enhance its usability and effectiveness.
6. The system should be able to function in a variety of indoor environments, such as shopping malls, airports, office buildings, and public transportation hubs, providing accurate positioning information regardless of the setting.
7. The solution can provide users with additional information, such as public facilities such as toilets and drinking rooms that are specially marked, to improve the overall user experience.
8. The system should be designed with an emphasis on accessibility, ensuring that it can be used by individuals with varying levels of technical proficiency, physical ability, or language skills.

1.4 Project planning

a. Preparation & Discussion

Following the direction of the team leader, group members convened regular meetings to share their perspectives and concepts, engage in discussion and

analysis, and ultimately arrive at several viable solutions. After a brief period of individual experimentation and investigation, the team collectively decided on a final feasible implementation strategy, which was subsequently refined and adjusted.

b. Development & Update

Weekly meetings were conducted, in which team members provided progress updates, resolved encountered challenges, shared their development experiences, and were assigned new responsibilities. Throughout this process, the team encountered several unanticipated obstacles, which were overcome through considerable effort and perseverance. Although a few issues remained unresolved, the primary objective of this project was successfully attained.

c. Testing & Summary

The prototype underwent practical testing and was documented, following which we fulfilled supplementary obligations such as creating a display poster, producing a demonstration video, and generating a project evaluation report.

2. Design concept comparison, development and screening

Our indoor positioning solution is intended to expedite and precisely facilitate users in locating their desired destinations, particularly in large, multi-story structures, such as searching for a classroom. We acknowledge that GPS signals are frequently unobtainable in indoor settings, necessitating the utilization of alternative positioning technologies to address this issue. Accordingly, we have evaluated four potential design alternatives:

1) WIFI Positioning.

General description:

WIFI signals are transmitted in the air at a known rate - the speed of light. The time between a signal being emitted from a hotspot (AP) or base station (STA) and being accepted by another AP or STA multiplied by the speed of light can be converted into distance. If there are two signal reception points in the area, two distances can be obtained, and then the position of the signal source can be calculated using the triangulation method[2].

Advantages:

Accurate positioning and fast information transmission speed.

As long as enough base stations are deployed, positioning services can be covered over a very large area. However, there is no need for such a large service range in our teaching buildings.

Disadvantages:

The deployment cost of WIFI base stations is relatively high.

WIFI positioning only works when the WIFI connection is opened, which could be unfriendly to certain users.

The location function is not available when the WiFi signal is between base station switches.

2) Visual recognition technology.

General description:

Visual recognition technology, A method or system that locates a monitoring target using a video surveillance device. The traditional surveillance system is only responsible for collecting images, and ultimately it is up to humans to obtain data from the images. Visual recognition technology can automatically obtain the location information of the collected person according to the collected information[3].

Advantages:

The positioning method of the present invention only needs to slightly modify the original monitoring equipment or only need to record the information of each monitoring equipment to realize the function of monitoring target positioning. Also the farther target can be seen through the monitoring equipment. Therefore, it has the characteristics of low cost, high precision and long distance measurement.

Disadvantages:

This technology may violate the user's privacy, because it involves monitoring, the user's personal information will be adopted, once the software is attacked, the user's information will be leaked, especially the right of portrait will be violated.

3) Bluetooth position.

General description:

Indoor location of users using Bluetooth Low Energy (BLE) beacons. BLE beacons are small, low-cost devices that transmit a unique identifier to a smartphone or other receiving device within range. The receiving device calculates its location by measuring the signal strength from multiple beacons and using trilateration. Compatible with Android and iOS devices, this indoor positioning solution is suitable for a wide range of users[4].

Advantages:

i. BLE beacons are energy efficient and can extend the battery life of the device.

ii. The deployment and management of the system is relatively simple, as it only requires the installation of beacons and the development of a compatible smartphone application.

iii. The technology is compatible with the vast majority of smartphones, increasing its potential user base.

iv. By adding or removing beacons, the system can be easily scaled up or down, allowing greater flexibility in coverage size and accuracy.

Disadvantages:

i. The accuracy of BLE-based positioning systems may be affected by signal interference and obstruction by physical obstacles such as walls and furniture.

ii. Positioning accuracy is usually lower than some other indoor positioning technologies, such as UWB, PDR.
iii. The user needs to enable Bluetooth on the smartphone for the system to work properly, which may affect the battery life of the device.
iv. Positioning accuracy may be affected by the density and placement of beacons, which requires careful planning during deployment.

4)PDR indoor positioning.

General description:

PDR (Pedestrian Dead Reckoning) indoor positioning is a technology that uses sensors, such as accelerometers, gyroscopes, and magnetometers, to track the movement of a person within an indoor environment. By measuring the changes in the person's movement, PDR can estimate their current location with a high degree of accuracy[1].

Advantages:

High accuracy: PDR provides accurate location tracking even in areas where GPS signals are weak or non-existent, such as indoor environments.

Low cost: PDR sensors are relatively inexpensive and can be easily integrated into mobile devices.

Real-time tracking: PDR provides real-time tracking of a person's location, making it useful for applications such as emergency response or asset tracking.

Disadvantages:

Limited coverage: PDR is limited to indoor environments and may not work well in areas with complex structures or obstacles.

Requires calibration: PDR sensors require calibration to ensure accurate location tracking, which can be time-consuming and may require specialized equipment.

Drift error: PDR is prone to drift error, which can accumulate over time and cause inaccuracies in location tracking.

Our team comprehensively discussed the advantages and disadvantages of the above four indoor positioning systems, taking into account factors such as accuracy, equipment cost, and implementation difficulty. After careful consideration, we decided to use the fourth design concept: PDR indoor positioning system. Therefore, we have decided to create a prototype of our indoor positioning product based on this design.

3. Prototyping and testing

3.1 Product Design

The design of our PDR location application represents an interdisciplinary effort with the overarching aim of delivering an integrated and optimal user experience. The app's UI design is centered on intuitiveness and ease-of-use, prioritizing accessibility and straightforwardness in the presentation of its features. The application's user interface incorporates an interactive indoor map feature, allowing

users to navigate indoor spaces seamlessly and establish their precise location within a building.

The PDR location app provides users with a range of comprehensive product features to enhance their experience. Among these features, the capability to customize the app to their individual stride length is a crucial one. This feature is particularly advantageous for users whose stride length deviates from the average person, enabling them to tailor the app to their specific physical characteristics.

An additional noteworthy functionality of the PDR location app is the capacity to rectify orientation. This feature enables users to modify the app's orientation and reconcile any inconsistencies between their bodily movement and the app's tracking. This feature is especially beneficial in scenarios where the user's physical movement does not precisely align with the app's tracking.

In addition, the app provides a visualization of the user's acceleration. This feature allows users to see a graphical representation of their acceleration and deceleration, which can be useful for tracking their progress over time and identifying areas for improvement in their walking or running technique. In general, our team has focused on delivering a smooth and intuitive user experience for the PDR location application through our product design. Our emphasis was primarily on the app's UI interface and its capability to provide accurate movement tracking and valuable feedback to users. Furthermore, we aimed to make the experience customizable to suit the unique needs of each user.

3.2 Logic Design

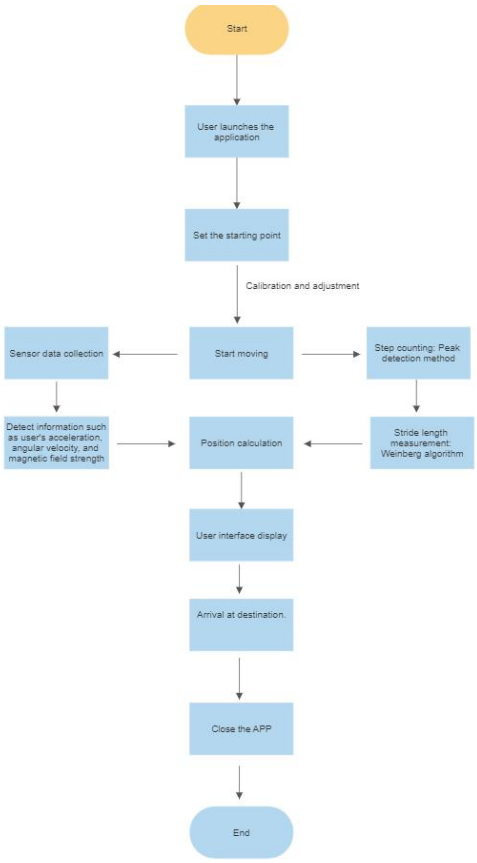


Figure 11. The logic diagram

3.3 Algorithms

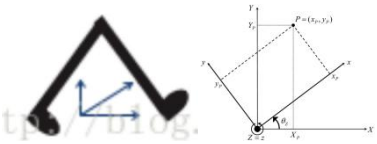
The implementation of PDR functionality primarily relies on three algorithms: Step Detection, Step Length Estimation, and Estimation of heading. The next step in the paper will provide a detailed explanation of these three algorithms.

3.3.1 Step Detection:

The acceleration sensor is equipped with three types of sensors: TYPE_ACCELEROMETER, TYPE_GRAVITY, and TYPE_LINEAR_ACCELERATION. The LINEAR_ACCELERATION sensor can detect acceleration values relative to the Earth but excludes the effects of gravity acceleration. Hence, we will use the output values of this sensor for PDR instead of the ACCELEROMETER sensor. We calculate acceleration values in two ways, considering only the acceleration values in the z'Oy' plane of the human coordinate system due to variations in body posture and grip. The zOy plane has Oz perpendicular to the ground and Oy tangent to the ground.

The gravity in each device direction is determined by the angle θ between the z'Oy' plane of the body coordinate system and the zOy plane. The acceleration value in the Oz direction (perpendicular to the ground) is more stable than that in the Oy

direction. Therefore, we will use the acceleration value in the Oz direction to estimate the step[5].



Foodstep analysis and acceleration in coordinate systems

Footstep detection algorithm: Step peak detection algorithm

Step peak detection algorithm is a method used to identify the peaks in acceleration signals that correspond to the start and end points of each step taken by a user. The algorithm involves setting a threshold on the acceleration signal and detecting the local maxima that exceed the threshold as potential step peaks. To filter out false positives, additional criteria such as minimum peak height and width are often applied[6]. The step peak detection algorithm is a critical component of pedestrian dead reckoning (PDR) systems that rely on inertial sensors to estimate the user's walking trajectory. Accurate and robust step peak detection is essential for reliable PDR navigation.

3.2.2 Step Length Estimation:

There are two methods for estimating step length in pedestrian dead reckoning: static and dynamic. The static method allows for the determination of step length based on a user's height. However, this method is not accurate when the user is moving quickly or slowly. Therefore, in our design, we use a dynamic method for step length estimation based on the motion acceleration: L = f(A), where A is the acceleration value determined by the equation. We use the Weinberg Method: this method determines the step length based on the difference between the maximum and minimum acceleration values for each step, as shown in the formula below:

L = $Step_size = k\sqrt[4]{A_{\max} - A_{\min}}$

where L is the step length, k is a scaling factor, Amax is the maximum acceleration value, and Amin is the minimum acceleration value[7].

3.3.3 Estimation of Heading:

There are various methods to calculate the motion direction of a device, such as using magnetic field sensors and gravity sensors to estimate the orientation of the device relative to the three coordinate axes placed on the Earth. However, due to the unstable gravity and magnetic fields at different locations, there can be significant tolerance in determining the orientation on the horizontal plane. Therefore, in our design, we use another method to estimate the device direction: the gyroscope sensor. This sensor measures angular momentum in the three body axes, which is used to predict the motion direction with low tolerance.

In our experiment, the Android phone that we used comes with a gyroscope and a magnetometer. The magnetometer is used to provide the user's orientation, while the gyroscope provides three parameters: pitch, roll, and yaw[8].

Magnetometer:

In the estimation of heading, the magnetometer plays a critical role in providing information about the user's orientation with respect to the Earth's magnetic field. By measuring the strength and direction of the magnetic field, the magnetometer can determine the user's heading and azimuth angle, which are essential for estimating the user's walking trajectory. The magnetometer is typically used in combination with other sensors, such as accelerometers and gyroscopes in our design, to improve the accuracy and robustness of the PDR system.

Gyroscope:

In the estimation of heading, the gyroscope plays a crucial role in providing information about the user's orientation and motion. By measuring the rate of rotation around the three axes of the device, the gyroscope can determine the user's pitch, roll, and yaw, which are essential for estimating the user's walking trajectory. The gyroscope data is integrated over time to estimate the change in orientation and displacement of the user.

Accelerometer:

Indeed, the accelerometer also plays a crucial role in the estimation of heading. In our experiment, the Android phone that we used also comes with an accelerometer sensor. The accelerometer measures the acceleration of the device along three axes, which can be used to determine the device's orientation relative to the Earth's gravitational field. By combining the accelerometer data with other sensor data, such as magnetometer and gyroscope data in our design, the estimation of heading can be improved, leading to more accurate and robust PDR positioning.

Codes of our algorithms:

```
private SensorEventListener listener1 = new SensorEventListener() {//Acceleration
        @Override
        public void onSensorChanged(SensorEvent event) {
            final float alpha = (float) 0.8;
            float[] gravity = new float[3];
            gravity[0] = alpha * gravity[0] + (1 - alpha) * event.values[0];
            gravity[1] = alpha * gravity[1] + (1 - alpha) * event.values[1];
            gravity[2] = alpha * gravity[2] + (1 - alpha) * event.values[2];
             float xValue = event.values[0] - gravity[0];//Removal of noise interference from gravitational acceleration
            float yValue = event.values[1] - gravity[1];
            float zValue = event.values[2] - gravity[2];
            float[] gravity1 = new float[3];
            gravity1 = event.values.clone();
            accelerometerReading[0] = event.values[0];//fed into the method for calculating R
            accelerometerReading[1] = event.values[1];
```

```
            accelerometerReading[2] = event.values[2];
            accelerometerReading = event.values.clone();
            acc = (float) Math.sqrt(Math.pow(gravity1[0], 2)
                    + Math.pow(gravity1[1], 2) + Math.pow(gravity1[2], 2));
            if (initialOrient == -1) {
                orient_utils_old= orient_utils = imu.oriAzimuth;;
                Log.i(TAG, "getRotateEndOrient: " + initialOrient);
            }
            orient_utils = imu.oriAzimuth;;
            if(Math.abs(orient_utils-orient_utils_old)>=25){
                orient_utils = imu.oriAzimuth;
            }else {
                orient_utils_old=imu.oriAzimuth;
                orient_utils=orient_utils_old;
            }
//            orient_utils = imu.oriAzimuth;
            float orient = orient_utils;
            View1.setText("orient: " + (orient + change_orient) % 360);
            new Thread(() -> {
                try {m2.updateDirection((orient + change_orient)%360);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }).start();
        }
```

The code uses the Accelerometer sensor.

The code segment defines a private SensorEventListener object named listener1 and implements the onSensorChanged method, which is called when the sensor values change. Inside the onSensorChanged method, a constant named alpha is defined, and a float array named gravity with a length of 3 is created. The code then calculates the accelerometer values without the influence of gravity using an exponential moving average. This is done by subtracting the previously calculated gravity values from the current accelerometer values.

Next, the code calculates the acceleration values in the x, y, and z directions based on the gravity-corrected accelerometer values. These values will be used for further calculations.

The code also creates another float array named gravity1 and retrieves the accelerometer values by cloning the event.values array, storing them in gravity1.

Furthermore, the accelerometer values are stored in a float array named accelerometerReading using the event.values.clone() method.

The code calculates the absolute value of acceleration (acc) by computing the modulus of the gravity1 array.

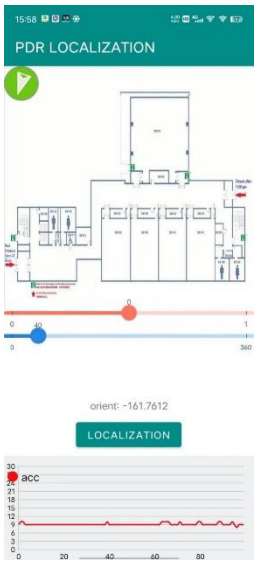
The code then determines whether to initialize orient_utils and orient_utils_old based on the value of initialOrient. If initialOrient is -1, orient_utils and orient_utils_old are set to the value of imu.oriAzimuth, and relevant information is logged.

Next, orient_utils is assigned the value of imu.oriAzimuth, and the code checks if the difference between orient_utils and orient_utils_old is greater than or equal to 25. If it is, orient_utils is set to imu.oriAzimuth. Otherwise, both orient_utils_old and orient_utils are set to the value of imu.oriAzimuth.

In the final part, the code calculates the value of orient (orient_utils plus change_orient modulo 360) and displays it in View1.

Additionally, in a new thread, the code calls the m2.updateDirection method and passes the calculated direction value (orient plus change_orient modulo 360).

User Interface (UI) of our design （Combination of code functions and instructions）：




```
private ArrayList<Line> lines = new ArrayList<Line>();
    private ArrayList<Line> lines1 = new ArrayList<Line>();
    public float[] gpdr(float len,float orient) throws Exception {
        float  [] xy=new float[2];
        x += (float) (len * Math.sin(Math.toRadians(orient)));
        y += -(float) (len * Math.cos(Math.toRadians(orient)));
        ViewPoint point = new ViewPoint();
        point.x = x;
        point.y = y;
        current.points.add(point);
        step_num+=1;

        xy[0]=x;
        xy[1]=y;
        invalidate();
        return xy;
    }
    @Override
    protected void onDraw(Canvas canvas) {
        super.onDraw(canvas);
        if (compass == null) {
            compass = getResources().getDrawable(R.drawable.t4);// Get the image resource of the current view
            compass.setBounds(0, 0, getWidth(), getHeight());//The location of the image resource in the view, here equivalent to filling the view
        }

        compass.draw(canvas);// Draw the rotated image on the view, i.e., keep the rotated look
        for (int i = 0; i < lines.size(); i++)
        {
            drawLine(canvas, lines.get(i));
```

```
        }
        drawLine(canvas, current);
    }
    public void updateDirection(float direction) {
        mDirection = direction;
        invalidate();// Refresh it and update the direction
    }
    private void drawLine(Canvas canvas, Line line)
    {
        for(int i = 0; i < line.points.size()-1; i++){
            float x = line.points.get(i).x;
            float y = line.points.get(i).y;
            float nextX = line.points.get(i+1).x;
            float nextY = line.points.get(i+1).y;
            paint.setColor(Color.BLUE);
            canvas.drawCircle(x, y, 15, paint);
        }
    }
    public class Line {
        ArrayList<ViewPoint> points = new ArrayList<>();
    }
    public class ViewPoint {
        float x;
        float y;
    }
    public void init(){
        ViewPoint point = new ViewPoint();
        point.x = 511;
        point.y = 492;
        current.points.add(point);
    }
    public void clear(){
        current.points.clear();
        paint.reset();
        //// Set the brush color to transparent and repaint
        paint.setXfermode(new PorterDuffXfermode(PorterDuff.Mode.CLEAR));
        paint.setColor(Color.TRANSPARENT);
        invalidate();
    }
    @Override
    public boolean onTouchEvent(MotionEvent event) {
        float X = event.getX();
        float Y = event.getY();
        current.points.clear();
```

```
        x= event.getX();
        y=event.getY();
        ViewPoint point = new ViewPoint();
        point.x = x;
        point.y = y;
        current.points.add(point);
        current.points.add(point);
        System.out.println(X);
        System.out.println(Y);
        System.out.println("test");
        invalidate();
        return true;
    }
    public void load1() {
        i=0;
        current.points.clear();
        x=511;
        y=492;
        step_num=0;
        invalidate();
    }
    public void begin1() {
        i=1;
        LS.clear();
//        String path= Environment.getExternalStorageDirectory().getAbsolutePath()+"/aaa/";
//        System.out.println("path1"+path);
    }
```

- The code defines two private ArrayLists of type "Line" named lines and lines1.
- The gpdr method calculates the new position (x, y) based on the given length and orientation. It updates the values of x and y and adds a new ViewPoint to the current Line's points. It also increments the step_num variable. The method returns an array of floats (xy) containing the updated x and y values.
- The onDraw method is responsible for drawing the compass image and the lines on the canvas. It first checks if the compass image (compass) is null and initializes it if necessary. Then it draws the compass image on the canvas. Next, it iterates over the lines ArrayList and calls the drawLine method to draw each line on the canvas. Finally, it calls the drawLine method to draw the current Line.
- The updateDirection method updates the value of mDirection and calls invalidate to refresh the view and update the direction.
- The drawLine method is called by onDraw to draw a line on the canvas. It iterates over the points of the given Line and draws circles at each point on the canvas.
- The code defines two inner classes: Line and ViewPoint. Line represents a collection of ViewPoints, and ViewPoint represents a point with x and y coordinates.

- The init method initializes the current Line with a default ViewPoint at coordinates (511, 492).
- The clear method clears the current Line's points, resets the paint, sets the paint color to transparent, and invalidates the view to repaint it.
- The onTouchEvent method is called when a touch event occurs on the view. It updates the current Line's points with the touch coordinates and adds the updated point to the current Line. It then calls invalidate to refresh the view.
- The load1 method clears the current Line's points, resets the x and y coordinates to default values (511, 492), sets the step_num to 0, and invalidates the view.
- The begin1 method sets the value of i to 1 and clears the LS ArrayList.

3.4 Testing and feedback

In terms of design requirement:
Sensors used in this project should be tested to make sure they reliability.
In terms of user needs:
Firstly, the positioning function should be tested, including whether there appear significant errors when user makes a turn, and whether the positioning accuracy is good enough to avoid misleading.
Secondly, the customization function should be tested, which refers to whether users could insert the map they need, and whether the step length could be adjusted with personalization.
Thirdly, the information diversification should be tested. This indicates that users should be able to obtain various information through this mobile application, including the position of toilets, drinking room, staircase, emergency exit etc.
Lastly, the accessibility should be tested. For the convenience of the most users, this positioning system should be accessible by various levels of devices or unsophisticated for different kinds of people.

3.4.1 Testing

In terms of design requirement:
Sensors used in this project should be tested to make sure they reliability.

In terms of user needs:
To begin with, the positioning functionality requires thorough testing to determine the presence of significant errors when users change direction and the sufficiency of positioning accuracy to prevent erroneous guidance. The testing procedure involves the following steps: after launching the application, the user sets a starting point in the building based on their current location. Subsequently, while moving in a straight line, all functions should be operational. Next, the user rotates 180

degrees, followed by 90 degrees, and finally, another 180 degrees. The results of the test indicate a marginal deviation of 10 degrees.

Secondly, it is necessary to test the customization functionality to assess whether users can personalize the step length. The results indicate that users can customize their step length according to their requirements, even while walking. The adjustment range accommodates the user's movement from standing still to running, with complete support.

Thirdly, it is imperative to evaluate the information diversification aspect of the mobile application. This feature allows users to access a range of information, such as the location of toilets, drinking rooms, staircases, emergency exits, among others. To test this functionality, users begin at a random point in the building and utilize the application to navigate to a classroom. After a few turnings, the user successfully reaches the destination, EE110, demonstrating the efficacy of the information diversification feature.

Lastly, it is vital to evaluate the accessibility of the positioning system to ensure that it can be used by individuals with varying levels of technical proficiency. To this end, random strangers were interviewed and queried about their ability to understand the application by simply observing it. The results are encouraging, as the majority of the participants recognized that the application was a positioning system, indicating that it is accessible to a diverse range of users, including those with limited technical experience.

3.4.2 User feedback

Once our product was finalised and all the features were working, we again brought in the students we had interviewed earlier to test our product. Here is their real feedback after using it:

Student A(Year 1 Domestic): I think it's great that it not only shows me where I am, but also gives me directions so I know where I should be going.
Student B(Year 1 From Indonesia): After trying this software, I think it's perfect for me. If I use this app in the science building, I won't miss some key corners and miss the right way when I'm looking for directions. I hope your software will be updated soon with a map of the Science Building.
Student C(Year 1 Domestic): I find it interesting that I can manually set different starting points and that I can also adjust my step length. This way no matter where in the school building I start I can locate myself in real time and find my destination precisely.

4. Conclusion

In summary, the PDR location application has been designed to provide users with precise and dependable indoor navigation. This has been achieved through the incorporation of a range of sensors, including the accelerometer, magnetometer, and gyroscope, which are utilized to estimate the user's walking trajectory and deliver real-time position updates. Additionally, the application boasts a user-friendly interface and customizable map options, which contribute to its accessibility to a diverse range of users.
Our design fosters principles of equality, diversity, and inclusion by offering a navigation solution that accommodates individuals of all physical abilities and mobility levels. Additionally, the application promotes environmental sustainability by decreasing the reliance on paper maps and printed directions.
Throughout the project, each team member made significant contributions to the success of our PDR location app. Heming was responsible for mapping, PDR algorithm calculation and code implementation, and error analysis. Yifei provided leadership by organizing meetings, making decisions, and overseeing the integration of all software and code. Yihan conducted user surveys, implemented step detection code, and helped with software and code integration. Qixiang was responsible for using the phone's sensors and capturing information, as well as producing videos and repairing code. Finally, Junjie was responsible for angle detection code implementation, poster production, software and field testing.
Through the effective utilization of our varied skill sets and collaborative efforts, we have successfully developed a PDR location application that is characterized by its accuracy, reliability, and user-friendliness. Our design ethos prioritizes principles of equality, diversity, and inclusion, resulting in an accessible navigation solution that caters to the needs of all users, irrespective of physical ability or mobility level. Furthermore, the application aligns with environmentally sustainable practices by reducing the dependence on printed maps and directions.
In order to advance the capabilities and effectiveness of our application, we propose that future research and development efforts prioritize the enhancement

of accuracy and robustness, particularly in regions with pronounced magnetic interference. Furthermore, we advocate for the exploration and incorporation of supplementary functionalities such as voice-activated navigation and augmented reality options, aimed at augmenting the user experience. Ultimately, our team takes pride in the accomplishment of our final design and the invaluable contributions of each team member.

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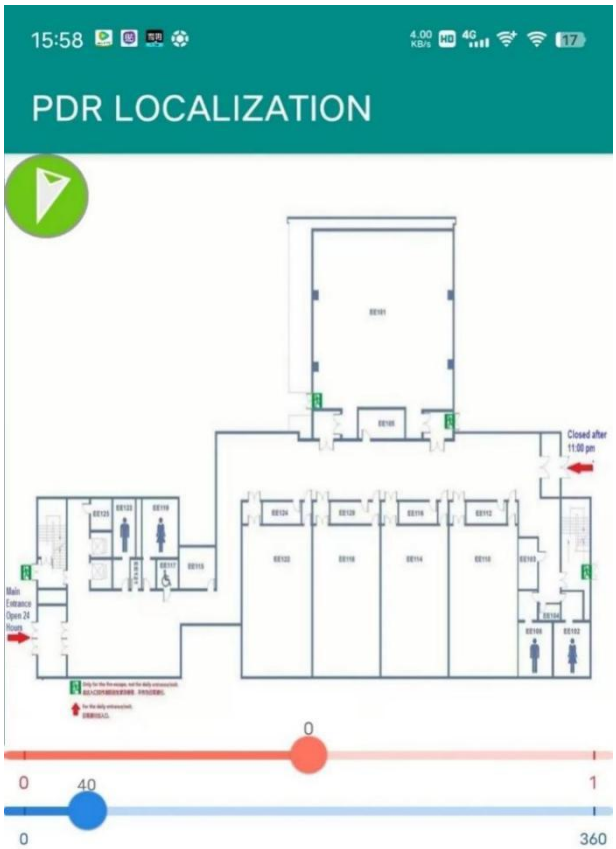
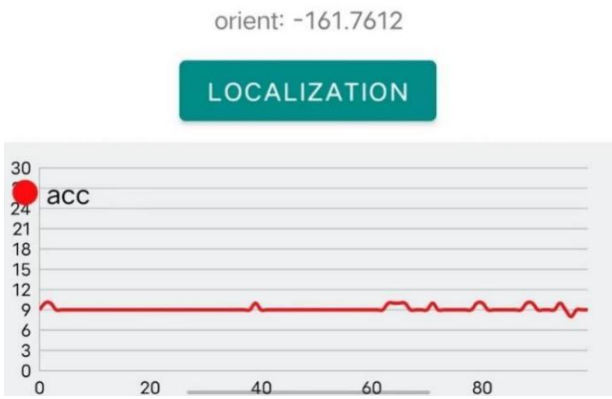
6. Appendices

Appendix A: Engineer Drawings

User Interface (UI) of our design:





Appendix B: Components and Price

Table : Components' name and price list

Name	Price(yuan)
IQOO9	4399
Magnetometer	0
Accelerometer	0
Gyroscope	0
Total	=4399

Appendix C: Individual Contribution Form

Student Name/ID	Contribution percentage	Brief description of individual contribution
Heming.Chen 2034634	20%	1. Be responsible for mapping 2. PDR algorithm calculation and code implementation. Error analysis
Yifei.Shen 2034340	20%	1.team leader, Organize meetings and make decisions. Push the project forward 2. Integration of all code and software generation 3. Report writing
Yihan.Zhang 2034010	20%	1. Be responsible for User survey 2. Step detection and code implementation 3. Integration of all code and software generation
Qixiang.Hu 2036499	20%	1. Be responsible for the use of sensors on the phone, and the invocation of captured information 2. Video producing 3. Code repair
Junjie.Qiu 1929268	20%	1. Be responsible for Angle detection and code implementation 2. poster producing 3. Software testing and Field testing.

Appendix D: Group Meeting Records

Date: 2nd April

Attendees: Qixiang.Hu, Junjie.Qiu, Yifei Shen, Yihan.Zhang, Heming Chen

Agenda: Discussion on the progress of the project and any issues faced

Meeting Record:

Qixiang.Hu reported that the open-source project code found on the internet did not work when trying to call the built-in accelerometer in the mobile phone. Junjie.Qiu also mentioned that the method which uses to insert gyro sensor data to the computer found online doesn't work well.

Yifei Shen raised concerns about difficulty in organizing team meetings and making decisions. Yihan.Zhang reported that no data could be collected after the questionnaire was made, and few people filled it out carefully.

Heming Chen shared that he had difficulty understanding the mapping process and how it is implemented in an indoor positioning system.

Action Items:

Qixiang.Hu and Junjie.Qiu agreed to work together and find an alternative solution to the problem of calling the built-in accelerometer in the mobile phone and inserting gyro sensor data.

Yifei Shen will take the lead in organizing team meetings and ensuring that decisions are made in a timely manner.

Yihan.Zhang will follow up with volunteers who have not yet filled out the questionnaire and encourage them to do so.

Heming Chen will schedule a meeting with the project manager to clarify any doubts regarding the mapping process in the indoor positioning system.

Date: 9th April

Attendees: Qixiang.Hu, Junjie.Qiu, Yifei Shen, Yihan.Zhang, Heming Chen

Agenda: Discussion on the progress of the project and any issues faced

Meeting Record:

Qixiang.Hu reported that calling two or more sensors simultaneously will cause the program to crash. Junjie.Qiu added that when the program calls the gyroscope sensor method and the accelerometer method at the same time, the gyroscope sensor data is erroneous.

Yifei Shen raised concerns about challenges with integrating all code and generating PDR (Pedestrian Dead Reckoning) location. Yihan.Zhang reported that he could not use "as" to call the sensor in the phone, and therefore, was unable to start the first step of software production.

Heming Chen shared that he encountered challenges while implementing the PDR algorithm, resulting in inaccurate initial results.

Action Items:

Qixiang.Hu and Junjie.Qiu agreed to work together and find a solution to the issue of calling multiple sensors simultaneously, and to fix the error in the gyroscope sensor data.

Yifei Shen will take the lead in integrating all code and generating PDR location.

Yihan.Zhang will explore alternative methods to call the sensor in the phone and start the first step of software production.

Heming Chen will work on improving the accuracy of the PDR algorithm.

Date: 16th April

Attendees: Qixiang.Hu, Junjie.Qiu, Yifei Shen, Yihan.Zhang, Heming Chen

Agenda: Discussion on the progress of the project and any issues faced

Meeting Record:

Qixiang.Hu reported that when trying to make a simple estimation of the obtained sensor output value, a significant error was found between the actual value and the theoretical value.

Junjie.Qiu mentioned that when testing the final product, it was found that when walking in a straight line, the positioning error was small, but when walking in a diagonal line, the distance displayed on the program changed more than the actual distance.

Yifei Shen raised concerns about struggles with report writing and organizing content.

Yihan.Zhang reported that when testing the software, the measured step size was always wrong, and in addition, the direction would be biased after accumulation. Most seriously, the software keeps flashing back on Huawei phones.

Heming Chen shared that during testing, localization estimation errors were found, especially when walking on diagonal paths.

Action Items:

Qixiang.Hu and Junjie.Qiu agreed to work together and find a solution to the significant error between the actual value and the theoretical value obtained from the sensors, and to improve the accuracy of diagonal path tracking.

Yifei Shen will take the lead in report writing and organizing content.

Yihan.Zhang will work on fixing the issues with step size and direction measurement, and optimizing the software for Huawei phones.

Heming Chen will work on improving the localization estimation accuracy, especially on diagonal paths.

Date: 23rd April

Attendees: Qixiang.Hu, Junjie.Qiu, Yifei Shen, Yihan.Zhang, Heming Chen

Agenda: Discussion on the progress of the project and any issues faced

Meeting Record:

Qixiang.Hu reported that the rotation of the direction could not be achieved by using only the values output of the accelerometer.

Junjie.Qiu mentioned that during the test, it was found that the preset step size caused great error in the actual test.

Yifei Shen raised concerns about difficulties in meeting project deadlines and staying on track.

Yihan.Zhang reported that in this test, the directional deviation has been greatly improved, but the step size problem still plagued the ranging.

Heming Chen shared that a discrepancy was found between the calculated and actual step size, leading to significant errors in positioning.

Action Items:

Qixiang.Hu and Junjie.Qiu agreed to explore alternative methods to achieve rotation of the direction and improve the accuracy of step size measurement.

Yifei Shen will take the lead in ensuring that project deadlines are met and the team stays on track.

Yihan.Zhang will work on improving the ranging accuracy by addressing the step size problem.

Heming Chen will work on improving the accuracy of positioning by resolving the discrepancy between the calculated and actual step size.

Date: 7th May

Attendees: Qixiang.Hu, Junjie.Qiu, Yifei Shen, Yihan.Zhang, Heming Chen

Agenda: Discussion on the progress of the project and any issues faced

Meeting Record:

Qixiang.Hu reported that when shooting a demonstration video, the camera cannot simultaneously balance the presenter's walking and the map on the phone screen.

Junjie.Qiu mentioned that the team had a disagreement over the presentation of the poster.

Yifei Shen raised concerns about challenges with finalizing the project and preparing for the presentation.

Yihan.Zhang reported that the team still had different opinions on the presentation video and the final report. The editing of the video was not organized enough, and the final rehearsal of the class presentation often went awry.

Heming Chen shared that the team was facing disagreements regarding the presentation and layout of mapping results.

Action Items:

Qixiang.Hu and Junjie.Qiu will work together to find a solution to balance the presenter's walking and the map on the phone screen when shooting the demonstration video.

Yifei Shen will take the lead in finalizing the project and preparing for the presentation.

Yihan.Zhang will work on organizing the video editing and coordinating the final rehearsal of the class presentation.

Heming Chen will work on resolving the disagreements within the team regarding the presentation and layout of mapping results.

Appendix E: Student's logbook

Student's logbook: Qixiang.Hu

Date	Problems encountered	Solutions
2th April	The open-source project code found on the internet does not work when trying to call the built-in accelerometer in the mobile phone.	After multiple attempts, the problem still exists. We searched for other open-source projects and learned about applications, and found that some of the code could run successfully, but some may encounter problems. We speculate that it may be due to different system versions.
9th April	Calling two or more sensors simultaneously will cause the program to crash.	We checked the code multiple times and found no errors. We accidentally ran it on another device and found that the problem was resolved. We speculate that the sensor of the first device has malfunctioned.
16th April	When trying to make a simple estimation of the obtained sensor output value, we found that there was a significant error between the actual value and the	We checked the output value of the sensor and found that the α， β， γ output of the gyroscope sensor had an offset that did not match the actual situation. We calibrated and

	theoretical value.	reset the parameters of the gyroscope sensor, temporarily resolving this issue.
23th April	The rotation of the direction could not be achieved by using only the values output of the accelerometer.	We consulted the professor and searched for relevant materials, and found that direction and rotation generally require the outputs of both accelerometer and magnetometer sensors.
7th May	When shooting a demonstration video, the camera cannot simultaneously balance the presenter's walking and the map on the phone screen.	We decided to shoot two videos. One to capture the demonstrator's walking state, and the other to be recorded on screen by the running device, then synchronously edit them into one video.

Student's logbook: Junjie.Qiu

Date	**Problems encountered**	**Solutions**
2th April	The method which uses to insert gyro sensor data to the computer found online doesn't work well	We read some Android studio manuals and improved the method on the web. This way, we can easily call the sensor's data in our program
9th April	When the program calls the gyroscope sensor method and the accelerometer method at the same time, the gyroscope sensor data error	We checked that both methods call the same variable, and we modified the gyroscope method
16th April	When testing the final product, we found that when we walked in a	We use a factor of less than 1 times the actual Angle on the sensor in the gyroscope

	straight line, the positioning error was small, but when we walked in a diagonal line, the distance displayed on the program changed more than we actually did	method. In the field test, the error is found to be reduced
23th April	During the test, we found that the preset step size caused great error in the actual test	The team member who is responsible for step size changed the step size in the program to be customizable
7th May	We had a disagreement over the presentation of the poster	In the final discussion, we decided to simplify the poster to make it easier for viewers to understand the whole project

Student's logbook: Yifei Shen

Date	**Problems encountered**	**Solutions**
2th April	Difficulty in organizing team meetings and making decisions	Scheduled regular team meetings and established a decision-making process to ensure everyone's input is taken into account before making any decisions
9th April	Challenges with integrating all code and generating PDR location	Collaborated with team members to identify the root cause of the issue and worked together to find a solution, which involved revising the code and testing
16th April	Struggles with report writing and organizing content	Developed an outline for the report and assigned specific sections to each team member to ensure that the report is well-organized and covers all necessary information

23th April	Difficulties in meeting project deadlines and staying on track	Re-prioritized tasks and established a clear timeline for completion to ensure that the project remained on track and all deadlines were met
7th May	Challenges with finalizing the project and preparing for presentation	Collaborated with team members to ensure that all aspects of the project were completed to a high standard and rehearsed the presentation to ensure it was polished and professional

Student's logbook: Yihan.Zhang

Date	**Problems encountered**	**Solutions**
2th April	No data could be collected after the questionnaire was made, and few people filled it out carefully.	Use the Wenjuanxing platform instead, and have group members send moments together to collect data.
9th April	Can not use as to call the sensor in the phone, can not start the first step of software production.	Seek answers on forums, browse open source projects for inspiration, and watch self-learning software tutorials. Finally, the algorithm is adjusted to gain access to the sensor.
16th April	When testing the software, we found that the measured step size was always wrong, and in addition, the direction would be biased after accumulation. Most seriously, the software keeps flashing back on Huawei phones.	We adjusted the constants in the algorithm formula and measured the field data again to adjust for deviations. For hardware problems, we switched to a vivo phone. The software flash back problem is solved.

23th April	In this test, the directional deviation has been greatly improved, but the step size problem still plagued our ranging.	Again, we improved the constants in the formula, and set a step size that can be adjusted to meet each person's different conditions.
7th May	We still have different opinions on the presentation video and the final report. The editing of the video was not organized enough, and the final rehearsal of the class presentation often went awry.	We repeatedly adjusted the order of the videos, repeatedly modified the explanation monologue of the videos and invited students who are good at editing videos to help give opinions. Finally, after numerous rehearsals for the product presentation, we finally expressed everything fluently and fully showed the results.

Student's logbook: Heming Chen

Date	**Problems encountered**	**Solutions**
2th April	Difficulty understanding the mapping process and how it is implemented in an indoor positioning system.	Reviewed and researched various resources on mapping techniques and their application to indoor positioning systems. Discuss with the team for a clearer understanding.
9th April	Challenges encountered while implementing the PDR algorithm, resulting in inaccurate initial results.	Reviewed the PDR algorithm and consulted with team members with expertise in sensor data. Necessary adjustments were made to the algorithm implementation to improve accuracy.
16th	During testing,	Revisit the PDR algorithm

April	localization estimation errors were found, especially when walking on diagonal paths.	and adjust the weighting factors for different walking directions to minimize errors. Further tests were performed to verify the improvements.
23th April	A discrepancy was found between the calculated and actual step size, leading to significant errors in positioning.	Working with the team, an adjustable step size feature was implemented in the system, allowing user customization to improve accuracy.
7th May	Facing disagreements within the team regarding the presentation and layout of mapping results.	Have a group discussion to resolve the issue and eventually agree on a simplified and user-friendly layout for the map presentation.